A Review of Multi-Page Visually Rich Document Understanding: Architectures, Techniques, and Persistent Gaps

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Degree: BACS (Honours)

Abstract

Multi-page visually rich document understanding (MP-VRDU) has emerged as a distinct research area driven by the prevalence of long-form documents such as contracts, scientific papers, and financial reports whose evidence is distributed across pages and modalities. While single-page document understanding has matured into a well-defined sub-field with established benchmarks, the multi-page setting introduces challenges that simple extension cannot address: total document length frequently exceeds backbone context windows, answer-bearing evidence is sparsely distributed, and cross-page structural relationships cannot be recovered from per-token layout signals alone. This review synthesises recent work in MP-VRDU, organising the field by how systems manage cross-page evidence rather than by OCR dependency, and proceeds from broad context to specific motivation. We first establish the broad landscape of four architectural families and three modality-specific challenges, then narrow to the retrieval-augmented and iterative agentic pipelines that drive the current frontier and the component-level training that strengthens them, and finally focus on the dataset ecosystem and three persistent gaps that current approaches have not resolved: data scarcity coupled with evaluation integrity concerns, the evidence pipeline bottleneck between coverage and fidelity, and cross-page structural reasoning that remains brittle across all current routes.

1 Introduction

Documents are the principal medium through which organisations, governments, and the scientific community record and exchange information. Contracts, financial reports, scientific papers, technical manuals, regulatory filings, and presentation decks are pervasive and structurally heterogeneous, interleaving dense text with tables, figures, diagrams, and complex layouts [1]. Manually extracting information from such documents is slow, expensive, and error-prone, motivating a sustained research effort to build automated document visual question answering systems that can answer natural-language questions over rendered document pages. For single-page inputs, this problem is largely solved. Document-specialised systems such as mPLUG-DocOwl1.5 [2] approach or exceed human performance on benchmarks covering scanned documents (DocVQA [3]), infographics (InfographicVQA [4]), and charts (ChartQA [5]), and recent frontier multimodal LLMs such as Gemini 3.1 Pro [6] can read a rendered page directly from pixels, handling single-page and short multi-page inputs reliably when input resolution is sufficient.

Real-world documents, however, almost always extend beyond a single page, and the evidence required for a realistic question is typically scattered across several pages or distributed across heterogeneous elements [1; 7]. One might assume that scaling a strong long-context multimodal LLM (MLLM) further would resolve this, but several structural obstacles prevent that resolution. First, input length scales with page count: each visually rich page contributes high-resolution figures, tables, and charts that quickly exceed the context window of existing language and vision-language models. The quadratic compute cost of self-attention [8] compounds this burden, making naive scaling infeasible at realistic document lengths. Second, the evidence required to answer a question is sparse and distributed across pages, requiring a system to navigate, retrieve, and reason across the entire document rather than read a single page in detail; even when the document fits within a backbone's nominal context window, long-context models attend poorly to evidence buried in the middle of long inputs [9], so expanding context length does not translate into proportional gains in answer quality. Third, explainability becomes more demanding, as models must not only produce an answer but also localise the pages or passages from which it was derived, a requirement that a monolithic forward pass through a long-context backbone does not naturally support. Together, these obstacles establish that MP-VRDU is not solved by frontier scale alone and motivate specialised architectures, evidence-selection strategies, and orchestration techniques. Our aim in this review is to identify the persistent gaps that motivate continued research in this area.

MP-VRDU has developed rapidly in recent years and has been treated only incidentally by prior surveys that remain anchored to the single-page setting or organise progress by OCR dependency [10; 11]. The remainder of this review proceeds from broad context to specific motivation. Sections 2–4 establish the broad landscape: the single-page substrate multi-page systems inherit, four architectural families organised through the lens of cross-page evidence management, and three modality-specific challenges that recur across them. Sections 5 and 6 narrow onto the retrieval-augmented and iterative agentic pipelines that now drive the frontier, surveying both their inference-time composition and the component-level training that strengthens them. Section 7 foregrounds the dataset ecosystem on which all of the above depends, drawing attention to the annotation and synthetic-generation pressures that shape what can be measured. Sections 8 and 9 converge on three persistent open gaps and the directions in which the field might address them.

2 Background: Single-Page VRDU

Unlike plain text, document pages are two-dimensional arrangements of text, tables, and figures, so position information must be encoded alongside content for a model to interpret them faithfully. Multi-page VRDU systems inherit most of their per-page mechanisms from the single-page literature, recently surveyed by Ding et al. [10]. We summarise the two organising axes that recur into the multi-page setting: the OCR-dependent versus OCR-free split at the architectural level (§2.1), and three orthogonal modality-handling choices that persist independently of that split (§2.2).

2.1 Architectural Families

The dominant axis at the single-page level distinguishes two architectural families. **OCR-dependent frameworks** preprocess the page with off-the-shelf optical character recognition (OCR) or PDF parsers, producing recognised text and bounding-box coordinates that are either fed into the language model prompt [12] or projected through an auxiliary multimodal encoder such as LayoutLMv3 [13; 14], a transformer pretrained to jointly embed text, layout, and image patches. These systems rely on external tools to capture structural information without extensive visual pretraining, but inherit cumulative parser errors, particularly on handwritten or low-quality scanned documents, and the low-resolution document images they typically consume limit visual expressiveness. **OCR-free frameworks** discard the parser and have a vision encoder read text directly from rendered page pixels [2], with the language model trained to recover text and layout through pretraining objectives such as text recognition, grounding, and captioning [15]. These frameworks enable end-to-end training and handle handwriting and complex layouts that OCR cannot, but accurate fine-grained text recognition demands high-resolution input and large-scale pretraining, which the visual compression and shape-adaptive cropping strategies discussed in §2.2 are designed to manage.

2.2 Multimodal Representation

Across both architectural families, three orthogonal design decisions govern how text, layout, and visual modalities are encoded, with a fourth cross-cutting choice for how they are fused. **Text** is recovered as OCR strings injected into the prompt [12], through OCR-augmented multimodal encoders that produce joint text-layout-vision embeddings [14; 16], or as a training objective for the vision encoder itself through text recognition and detection tasks [17]. **Layout** is encoded as 2D positional features attached to text tokens [18; 16]; as verbalised bounding boxes or markdown-style formatting in the prompt [19], where coordinates such as `[x=100,y=200]` are interleaved with text; as specialised layout tokens that share positional information with their associated text [20]; or through supervised pretraining tasks such as visual text grounding and table reconstruction [21]. **Visual tokens** are managed through low-resolution patch embeddings [22], high-resolution shape-adaptive cropping that divides oversized pages into fixed-size sub-images [23; 2], or compression via resamplers, Q-Formers (small sets of learned query tokens that compress a long visual sequence), and convolutional reducers that cut the visual token count

Model	Venue	OCR	Vision Encoder	LLM Backbone	Mod.	Reasoning	Search	Agentic
Page-to-Document Encoding Models								
Arctic-TILT (2025)	ACL	Yes	U-Net	T5-Large	T, V, L	-	-	-
GRAM (2024)	CVPR	Yes	DocFormerv2	DocFormerv2	T, V, L	-	-	-
Hi-VT5 (2023)	PR	Yes	DiT	T5-Base	T, V, L	-	-	-
RM-T5 (2024)	DAS	Yes	DiT	T5-Base	T, V, L	-	-	-
Backbone-Centric MLLM Adaptation								
DocR1 (2026)	AAAI	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	CoT, HD	-	-
Leopard (2024)	TMLR	No	SigLIP-SO400M	LLaMa-3.1-8B, Mistral-7B	V	CoT	-	-
mPLUG-DocOwl2 (2025)	ACL	No	ViT-L/14	mPLUG-Owl2	V	-	-	-
DocSLM (2025)	CVPR	Yes	SigLIP2	Qwen2.5-1.5B	T, V, L	-	-	SC
InstructDoc (2024)	AAAI	Yes	CLIP	FlanT5, BLIP2	T, V, L	-	-	-
LayTokenLLM (2025)	CVPR	Yes	-	LLaMA3-8B, Qwen1.5-7B	T, L	-	-	-
CoR (2026)	preprint	Yes*	Qwen2.5-VL	Qwen2.5-VL-7B	T, V	CoT, PE, HD	-	-
Docopilot (2025)	CVPR	No*	InternViT-300M	Multiple (Intern)	V	-	-	-
Texthawk2 (2024)	preprint	No*	SigLIP-SO400M	Qwen2-7B-Instruct	V	-	-	-
Retriever-Generator Pipeline								
AVIR (2025)	ACM	No	Pix2Struct	Qwen2.5-VL-3B-AWQ	V	HD	D	1A
DFVC (2026)	MDPI	No	VisRAG, ColQwen	Qwen2-VL-2B-Instruct	V	-	D	-
M3DocRAG (2024)	preprint	No	ColPali, ColQwen	Multiple (Qwen/Intern/Idefics)	V	-	D	1A
MoLoRAG (2025)	EMNLP	No	ColPali	Multiple (Qwen/DS/LLaVA)	V	-	D, IR, Nav	1A
Self-Attention Scoring (2024)	ICDAR	No	Pix2Struct	Pix2Struct	V	-	D	1A
CREAM (2024)	ACM	Yes	Pix2Struct	LLaMA2-7B	T, V	HD	J	1A
KGP (2024)	AAAI	Yes	-	Multiple	T, L	-	S, IR, Nav	1A
MHier-RAG (2025)	preprint	Yes	Qwen-VL-Plus	Multiple (Qwen/DS/GPT)	T, V, L	CoT, HD	J	1A
MLDocRAG (2026)	preprint	Yes	-	Multiple (Qwen)	T, V, L	CoT, HD	D	1A
MultiDocFusion (2025)	ACL	Yes	-	Multiple (Qwen/LLaMa/Mistral)	T, V, L	HD	D	1A
M2RAG (2025)	NeurIPS	Yes*	VisRAG-Ret	Multiple (Qwen)	T, V	-	J	3A
MDocAgent (2025)	preprint	Yes*	ColPali, ColQwen2	Multiple (Llama/Qwen)	T, V	-	J	5A
PDF-WuKong (2024)	preprint	Yes*	IXC2-VL-4KHD	IXC2-VL-4KHD	T, V	-	J	1A
RAG-DocVQA (2025)	ICDAR	Yes*	DiT, Pix2Struct	Qwen2.5-VL-7B-Instruct	T, V, L	-	D	1A
VDocRAG (2025)	CVPR	No*	Phi3V-4.2B	Phi3V-4.2B	V	-	D	1A
Agentic Pipeline								
Doc-V* (2026)	preprint	No	Qwen2.5-VL	Qwen2.5-VL-7B-Instruct	V	ReAct, HD	D, IR, QR, Nav	1A
DocReact (2025)	ACL	No	ColPali, VisRAG	GPT-4o	V	ReAct, QD	D, IR, QR	1A
MACT (2025)	preprint	No	Multiple	Multiple (Qwen/Intern)	V	ReAct, PE	D	4A, SC, SV
DocLens (2025)	preprint	Yes	–	Multiple (GPT/Gemini/Claude)	T, V, L	ReAct	J	4A, SA
DocAgent (2025)	EMNLP	Yes*	–	Multiple (GPT/Gemini/Claude)	T, V, L	ReAct, HD	J, IR, Nav	3A
DocDancer (2026)	preprint	Yes*	Qwen3-VL-235B	Qwen3-4B/30B-A3B	T, V, L	ReAct	J, IR, QR	1A
SimpleDoc (2025)	EMNLP	Yes*	ColQwen-2.5	Multiple (Qwen)	T, V	CoT	J, IR, QR	1A, SR, SV
ViDoRAG (2025)	EMNLP	Yes*	NV-Embed-V2, ColQwen2	Multiple (Qwen/LLaMa/GPT)	T, V	ReAct, HD	J, IR	3A, SR, SV

Table 1: Survey of MP-VRDU models grouped by architecture. T = text; V = visual; L = layout. CoT = Chain of Thought; ReAct = Reason + Act; PE = Plan-Execute; QD = Question Decomposition; HD = Hierarchical / Coarse-to-Fine. S = Sparse; D = Dense; J = Joint; IR = Iterative Retrieval; QR = Query Reformulation; Nav = Navigation. nA = n agents; SR = Self-Refine; SV = Self-Verify; SC = Self-Consistency; SA = Sampling-Adjudication. OCR column: Yes* = not strictly OCR-dependent but incorporates OCR as an auxiliary framework component; No* = OCR-derived supervision is used during training but no OCR is used at inference. '–' indicates not reported / not applicable.

by an order of magnitude [24; 25; 21]. **Modality fusion** proceeds through dedicated neural encoders such as LayoutLMv3 that produce joint multimodal embeddings [13; 14], task-oriented objectives that bind modalities through shared supervision [2; 17], or prompt-based combinations that interleave layout descriptions with text and images at the language model input. Single-page systems make these choices assuming the entire document fits in a single backbone input; the multi-page setting reformulates each into a question of which page or region to attend to next, and fusion becomes a question of which evidence to combine after selective inspection.

3 Multi-Page Architectural Landscape

The single-page literature is naturally organised by OCR dependency, but this axis is less informative in the multi-page setting, as OCR-using systems often treat extracted text as one component in a multimodal pipeline [52; 55] while OCR-free systems are frequently pretrained or aligned with OCR-derived supervision [47; 2]. The defining challenge in MP-VRDU is instead how a system manages document-scale evidence: which information to retain, which to discard, and how the surviving information should interact across pages. We therefore organise multi-page frameworks primarily by architectural trend, with OCR dependency as a secondary descriptor. Through this evidence-management lens, the field has shifted from encoding all pages jointly toward increasingly selective and iterative document-level handling, producing four recurring families: (i) *page-to-document encoders* that fuse all pages in one forward pass with explicit cross-page mechanisms; (ii) *backbone-centric adaptation* that absorbs cross-page information inside a long-context MLLM; (iii) *retriever-generator pipelines* that select evidence before reading; and (iv) *agentic pipelines* that iteratively decide what to inspect. This is reflected in Table 1, which summarises

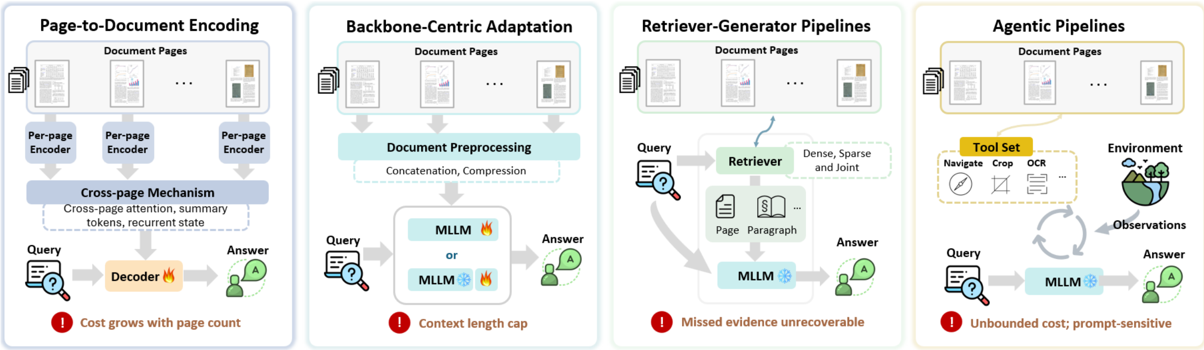

Figure 1: Illustration of MP-VRDU Architectures

specific model details, and in Figure 1, which illustrates the architectural differences between them.

3.1 Page-to-Document Encoding Models

Page-to-document encoding models process each page through a shared document encoder that fuses OCR text, layout, and visual features at the page level, then connect per-page representations through a document-level exchange mechanism within a single forward pass. The mechanism may take the form of per-page summary tokens [1], document-global tokens with local-global attention [7], recurrent memory states passed across pages [27], or sparse cross-page attention over long OCR sequences [26]. By preserving page identity, local layout, and reading order in the architecture itself, this design provides a strong inductive bias for multi-page structure and is well suited to documents with regular templates such as multi-page forms and business reports [1; 26]. The principal limitation is intrinsic to the design: every page is still encoded jointly in a single forward pass, and the per-page compression step that enables document-level exchange discards fine-grained evidence as inputs scale. As the context windows of pretrained MLLMs have widened, the motivation for a dedicated cross-page encoder has weakened, and subsequent families absorb cross-page information into the backbone or select evidence before reading.

3.2 Backbone-Centric Adaptation

Backbone-centric adaptation removes the dedicated cross-page encoder entirely and processes the document inside a pretrained MLLM whose context window absorbs cross-page information directly. Adaptation strategies span a spectrum of training intensity. At the lightweight end, *projectors* that map per-modality features into the backbone's token space [22] and *adapters* that insert small trainable modules into a frozen backbone for parameter-efficient fine-tuning [20] keep most weights fixed; visual-token compression further constrains token growth on long inputs to keep multi-page documents within the context budget [24]. At the heavier end, the backbone itself is unfrozen for continued pretraining or instruction tuning on document corpora, trading parameter efficiency for stronger document-specific grounding [32], with long-context tuning further improving performance on documents of moderate length [2; 25]. The family inherits the foundation model's instruction-following and reasoning, avoids the engineering overhead of external retrieval or orchestration, and remains parameter-efficient to train. The cost of this design is that performance degrades on longer documents [29]. Reliable text recovery depends on high-resolution input that aggressive token compression readily compromises, and document structure such as section hierarchy and figure-caption linkage must be recovered implicitly from pixels rather than supplied through dedicated tokens [30; 32]. Combined with the attention-dilution problem noted in §1, these limitations motivate subsequent families that move evidence selection out of the backbone altogether.

3.3 Retriever-Generator Pipelines

Retriever-generator pipelines decouple evidence selection from answer generation. A retrieval module returns a compact subset of OCR chunks, pages, visual patches, multimodal elements, or graph nodes, and a generator answers conditioned only on those units [35; 47]. The design adapts the retrieval-augmented generation paradigm from text [56; 57] to visually rich inputs, with the retriever instantiated as a sparse lexical scorer, a dense embedding model, or a structure-aware index, and the generator typically a pretrained MLLM. Inference cost depends on the retrieved content rather than document length, allowing the architecture to scale to documents that exceed any current MLLM context window. Modular decomposition lets retriever and generator be swapped independently, the retrieved evidence provides a partial audit trail for the answer, and the same interface extends to multi-document corpora without architectural change. Structured domains with reliable retrieval anchors such as scientific papers and technical manuals benefit most [42], and page-level visual retrieval preserves figures, tables, and layout that OCR-only pipelines lose [47; 45]. The principal limitation is the single-pass control flow: it assumes one retrieval step can surface all evidence required for the answer, evidence missed at retrieval is unrecoverable at generation, and broadening the retrieved set introduces noise and attention dilution. Performance is bounded by parser and embedding quality, and per-passage similarity scoring weakens when answers require aggregating weakly relevant pages or following multi-hop cross-references [39; 36], motivating the replacement of the fixed pipeline with adaptive, multi-step reasoning.

3.4 Agentic Pipelines

Agentic pipelines replace the fixed retrieve-then-generate control flow with iterative evidence acquisition. Drawing on the broader agentic-LLM framing [58; 59], we class as agentic any MP-VRDU system that performs multi-pass evidence acquisition or distributes work across specialised agents whose call sequence adapts at inference; multi-component systems with predetermined execution order, even when described as "agents," remain within the retriever-generator family. One or more *controllers* (MLLMs that decide what to inspect next from intermediate observations) issue tool calls for retrieval, navigation, cropping, or OCR [60; 52]. Roles such as retrieval, inspection, synthesis, and adjudication may be distributed across agents, each backed by a different backbone selected for its subtask [44] rather than tied to a single generator. This design lifts the single-pass bottleneck, allows compute to scale with question difficulty rather than document length, and renders the trajectory itself an inspectable audit trail. The family is well suited to questions whose evidence is distributed across pages or modalities and to multi-hop questions that benefit from query reformulation between steps [55; 52], as well as visually heterogeneous documents whose text, tables, and figures benefit from tool-specific inspection [44; 51]. The flexibility carries four costs. The variable-length action sequence makes inference cost difficult to bound before execution. Controllers remain sensitive to prompt phrasing and may commit early to wrong evidence with silent failure. Reproducibility is complicated by stochastic generation. And the pipeline inherits the recall ceiling of every retriever, parser, or OCR module it calls.

4 Multi-Page Multimodal Representation

While Section 3 categorises MP-VRDU systems by where cross-page evidence is processed, this section analyses the problem from a modality perspective. Single-page VRDU already handles text, vision, and layout through the well-established mechanisms outlined in Section 2. The multi-page setting introduces a shared pressure: the full document often exceeds the backbone's context window, and even when it fits, the answer occupies only a small fraction of a long input where attention is diluted and key evidence is easily lost in the middle [9]. This forces each modality to trade page *coverage* (how much of the document is available to the model) against reading *fidelity* (how accurately text and visual content are recovered from each page), with the resolutions differing across text, vision, and structure.

4.1 Text Modality

Text is the most information-dense modality in MP-VRDU but also the most heterogeneous in volume across pages and the most exposed to errors that compound when extracted across tens or hundreds of pages. Building on the OCR-extracted versus pixel-recovered distinction of §2.1, three patterns have emerged for handling text at document scale, each correlated with a different architectural family. **Per-page aggregation** encodes each page in detail and combines per-page representations into compact cross-page summaries through attention or memory tokens, typical of page-to-document encoders such as Hi-VT5 [1], GRAM [7], and Arctic-TILT [26]. **Whole-document compression** retains all pages within a single backbone context by aggressively reducing per-page text tokens through resamplers, layout-aware compressors, or token-reduction modules, typical of long-context backbone-centric MLLMs such as mPLUG-DocOwl2 [24] and TextHawk2 [25]. **Retrieve-then-read** abandons full-document encoding entirely, indexing text at chunk or page level and passing only a retrieved subset to the reader, characteristic of retriever-generator and agentic systems such as CREAM [38], MultiDocFusion [42], DocAgent [52], and MDocAgent [44]; agentic variants extend this into multiple selection-reading rounds. Each pattern carries a different risk: per-page aggregation discards token-level detail at the summary step, whole-document compression sacrifices fine-grained reading on individual pages, and retrieve-then-read makes evidence missed at retrieval unrecoverable at the reading stage. The last two routes also depend on document-level structure to decide which pages or sections to read in detail, an issue we return to in §4.3.

4.2 Visual Modality

Visual content in MP-VRDU spans text-bearing structures such as dense paragraphs and tables to non-textual evidence such as charts, figures, and handwriting that OCR cannot recover. Visual encoding faces a sharper version of the coverage-fidelity tension than text, since token cost scales jointly with page count and per-page resolution while the detail required to read charts, table cells, and handwriting cannot survive aggressive downsampling. Three patterns have emerged. **Aggressive token compression** keeps every page available in context but reduces per-page visual tokens through resamplers or convolutional reducers, as in mPLUG-DocOwl2 [24] and TextHawk2 [25]; this preserves coverage at the cost of per-page detail, and is often paired with OCR text to recover fine reading where the downsampled image is insufficient. **Selective high-resolution reading** restricts expensive encoding to a small subset of pages surfaced by retrieval or controller-driven inspection, sending those high-fidelity tokens into the reader while leaving other pages unread, as in Doc-V* [60] and SimpleDoc [54]. **Retrieval-only visual embeddings** decouple encoding from reading altogether: page images are encoded purely to produce retrieval embeddings, and detailed reading of selected pages is handed to a separate text or vision module, as in M3DocRAG [35] and VDocRAG [47]. The patterns combine in practice: several systems pair low-resolution thumbnails for navigation with selective high-resolution inspection of pages a retriever or tool call surfaces [60; 54]. The failure modes differ: aggressive compression loses fine text and small visual elements; selective high-resolution reading inherits the recall ceiling of whichever tool selects the pages; and retrieval-only visual embeddings may rank layout-similar pages without confirming that the answer-bearing region is present.

4.3 Cross-Page Structure

Within-page mechanisms inherited from single-page VRDU, including 2D positional encodings, bounding-box coordinates, and reading-order signals attached to individual tokens, describe only per-page geometry and do not extend to relationships between sections, pages, or references. The structural problem distinctive to MP-VRDU is *cross-page* structure: section hierarchies that span multiple pages, information referenced or continued across page boundaries, and figure-to-text or table-to-caption linkages that no longer fit within a single page's view. Two routes address it, with the second now dominant. Some backbone-centric systems train the model to predict cross-page structure end-to-end through multi-page parsing or document-level pretraining objectives [24; 32], but pretraining supervision at document scale remains scarce and expensive, limiting how far this approach can reach in practice. The dominant route exposes parsed structural information to the model directly, and three patterns recur within it. **Structure-aware chunking** uses a parser to extract section trees and heading hierarchies, then segments documents along semantic boundaries so retrieved units respect section scope rather than sliding-window cuts, as in MultiDocFusion [42]. **Hierarchical retrieval indices** route queries first to coarse units such as sections or chapters and then to fine-grained chunks within the selected branch, reducing irrelevant pages while preserving cross-page context, as in MHier-RAG [40] and MLDocRAG [41]. **Graph-based traversal** converts headings, entities, and cross-references into a knowledge graph whose typed edges encode relations such as "contains" or "refers to," allowing controllers to follow these relations across pages rather than relying on dense similarity alone, as in KGP [39]. All three depend on parser quality: errors in heading detection, table boundary segmentation, or reading-order extraction propagate directly into every downstream retrieval and navigation step, and reliable parsers themselves remain tuned for single-page or template-bound documents rather than the heterogeneous long-form documents on which MP-VRDU is evaluated.

5 Inference-Time Adaptation

Having organised MP-VRDU by architecture (§3) and modality (§4), we narrow to inference-time composition, where retrieval-augmented and iterative agentic pipelines now drive the current frontier. Inference-time adaptation refers to strategies that improve MP-VRDU performance through changes to the inference pipeline rather than to model parameters: a frozen backbone is paired with off-the-shelf retrievers, parsers, reasoning loops, or multi-agent orchestrations, with no gradient updates required. This approach suits MP-VRDU settings where supervised data is scarce, where backbones must remain exchangeable, or where rapid adaptation to new document types is needed. We organise these strategies along the inference pipeline: evidence selection through retrieval and navigation (§5.1), reasoning over the assembled evidence (§5.2), and coordination across multiple frozen components (§5.3).

5.1 Retrieval and Navigation

Inference-time retrieval selects a compact subset of pages, chunks, or graph nodes for the reader using off-the-shelf encoders, structural parsers, or LLM-based rerankers, without fine-tuning the retriever on document data. Retrieval quality bounds every downstream reasoning step, so the choice of signal dominates system behaviour on long documents. Two broad families have emerged, distinguished by whether passages are scored in isolation or navigated through explicit relations.

Similarity-based retrieval. Similarity-based methods score each passage against the query independently, using sparse lexical signals, dense embeddings, or combinations of both. The dominant dense signal in MP-VRDU is vision-language late-interaction encoding such as ColPali [61], which extends the late-interaction paradigm of ColBERT [62] to page images. It produces token-level query vectors and patch-level page embeddings over the rendered page, and scores a query-page pair by taking, for each query token, the maximum cosine similarity over all page patches and summing the results. This *MaxSim* aggregation preserves fine-grained match between specific query terms and specific page regions, critical when a single question token may need to align with one figure caption, table cell, or paragraph buried in a long document. M3DocRAG [35] retrieves at chunk and page granularity with this signal alone; AVIR [33] adds an adaptive selector that scales the number of returned pages with the score distribution rather than using a static top-K. Joint variants combine signals at the reranking stage: CREAM [38] pairs dense retrieval with iterative LLM-based reranking, SimpleDoc [54] combines ColPali embeddings with LLM-generated summary reranking under persistent working memory that refines the joint score across rounds, and MDocAgent [44] runs parallel text and image retrieval tracks aggregated by a downstream agent. Independent scoring handles paraphrase and cross-modal queries well, but cannot follow cross-page references or multi-hop dependencies that no single passage carries.

Structure-based retrieval. Structure-based methods exploit a parsed representation of the document to make passage relations first-class objects. Tree-based variants such as MultiDocFusion [42] and DocAgent [52] extract section hierarchies and outlines and group passages along semantic boundaries so retrieved chunks preserve parent context, while MHier-RAG [40] extends this by routing queries through dual in-page and cross-page indices with LLM-based reranking across both. Graph-based variants treat headings, entities, and cross-references as graph nodes with typed edges, allowing controllers to traverse from one passage to related ones rather than relying on similarity alone. KGP [39] uses an instruction-tuned LLM as a traverser over a knowledge graph of passages and structural nodes such as pages, paragraphs, and tables; MoLoRAG [36] and MLDocRAG [41] extend this with semantic-logical edge scoring and per-query graph construction respectively. Structure-based methods recover references that similarity scoring misses, at the cost of construction overhead, dependence on parser quality, and traversal cost that grows with hop count.

5.2 Inference-Time Reasoning Strategies

Inference-time reasoning strategies govern how a frozen language or vision-language model accesses, updates, and reasons over evidence during decoding, without modifying model parameters. In MP-VRDU, these strategies can be broadly categorised by whether evidence acquisition remains static or becomes dynamically interactive during inference. **Closed-buffer reasoning** operates over a fixed evidence set assembled before generation begins, whereas **tool-interactive reasoning** allows the model to iteratively acquire additional evidence through retrieval or page-level actions during the reasoning process.

Closed-buffer reasoning. Closed-buffer reasoning performs multi-step reasoning over a fixed evidence buffer assembled by a prior retrieval step or by a compressed document representation. Chain-of-thought prompting [63] is the canonical instantiation, eliciting intermediate reasoning steps that integrate textual, tabular, and visual evidence before producing the final answer. MHier-RAG [40] applies structured chain-of-thought prompts over hierarchical retrieval outputs, although a single reasoning chain remains vulnerable to compounding errors on dense layouts. SimpleDoc [54] adds self-refinement and self-verification prompts that emit revised queries when the assembled evidence is judged insufficient, but the loop terminates on the backbone's own judgement and inherits its blind spots. Closed-buffer reasoning is fundamentally bounded by retrieval completeness: evidence absent from the buffer cannot be recovered by subsequent reasoning or verification, regardless of how many self-checks are applied.

Tool-interactive reasoning. Tool-interactive reasoning interleaves reasoning and evidence acquisition during inference, allowing newly observed pages to guide subsequent reasoning steps. The ReAct paradigm [64], in which the model alternates intermediate reasoning traces with tool calls whose outputs feed the next reasoning stage, is the canonical instantiation. Doc-React [49] instantiates this with flat retrieval, contrastively-guided sub-query refinement, and entropy-based stopping, although each iteration still selects from a single retriever's ranked list and recovery from a poor encoder match relies on subsequent query rewrites. Outline-driven variants combine ReAct with the structured navigation of §5.1: DocAgent [52] traverses a tree-formatted XML outline through section-level search and fetch tools chosen turn-by-turn by an actor agent, inheriting the parser's outline quality and degrading on documents without exploitable structure. A complementary refinement is coarse-to-fine visual reading, where fetched pages are inspected at multiple resolutions (e.g., thumbnails for navigation and high-resolution encoding for inspection) so that expensive encoding is reserved for regions surfaced by the coarse pass. This concentrates compute on task-relevant regions but makes reasoning conditional on the localiser, since the fine pass cannot recover when the coarse pass misroutes. Adaptive evidence acquisition improves localisation and supports more exploratory multi-page reasoning, but introduces three costs: tool latency that grows with rounds, sensitivity to prompt design and to early action choices that subsequent steps cannot recover from, and attention dilution within fetched pages at their reading resolution.

5.3 Agentic Orchestration

Retriever-generator and agentic pipelines often distribute a single MP-VRDU question across multiple specialised language or vision-language model calls, each handling a particular modality, reasoning step, or verification role. The frozen backbones are not modified; coordination is achieved through prompt design and call ordering. The two families correspond to whether the trajectory of calls is fixed at design time or adapts to the question at inference.

Fixed-trajectory pipelines. Fixed-trajectory pipelines follow a control flow set in advance: which calls are made, in what order, and how their outputs are combined are all determined at design time. Modality-partitioned variants dispatch readers by input type and aggregate their outputs through a final synthesis step. MDocAgent [44] runs five role-partitioned agents (general context, critical evidence, text, image, and synthesis) whose outputs are fused by the final summariser. M2RAG [43] splits work between a text filter, a visual extractor, and a modal fuser that applies a confidence-conditional rule, falling back to one modality when the other reports no information and fusing only when both contribute. Such pipelines improve coverage relative to single-agent baselines, but lack an error-correction stage: a wrong choice of routing or ordering cannot be revised mid-execution, and an incorrect output from one branch can dominate aggregation when the other branch lacks counter-evidence.

Adaptive-trajectory pipelines. Adaptive-trajectory pipelines treat the call sequence itself as a variable: which agents are invoked and how many rounds the loop runs are decided per question. Loop-based variants such as ViDoRAG [55] cycle a seeker that retrieves coarse candidates, an inspector that reviews them at fine resolution and emits reflective feedback, and an answerer that performs a consistency check, with per-query bounds learned from the similarity distribution rather than a static top-K. Proposer-verifier variants insert an explicit verification stage between candidate generation and final answer: DocLens [51] samples multiple candidates and routes them through an adjudicator that compares reasoning paths and selects the most consistent conclusion rather than fusing them unconditionally, while DocAgent [52] pairs an actor with a reviewer that cross-checks through complementary tools and writes disagreements into a shared reflection memory that updates the actor's behaviour on subsequent rounds. MACT [50] decomposes the loop further into planning, execution, judgement, and answer agents, separating the judge from the corrector to avoid the same agent excusing its own errors. Adaptive-trajectory pipelines improve the accuracy-latency trade-off, but inference cost remains hard to bound before execution and the verifier typically draws on the same backbone family as the proposer, so shared blind spots persist when both agents fail in the same way.

6 Training Strategies

The inference-time strategies of Section 5 hold model parameters fixed, which keeps backbones exchangeable and avoids expensive supervision but leaves system behaviour bound to what off-the-shelf encoders, prompts, and orchestration can achieve. Many MP-VRDU systems therefore extend these pipelines by training the very components that drive the retrieval and iterative-orchestration emphasis of §5: replacing brittle retrieval signals with learned ones, prompted reasoning with internalised reasoning policies, and prompt-only orchestration with distilled or reinforcement-learned trajectories. These training entry points (retriever, reasoner, and orchestration policy) mirror the inference-time decomposition, with the rest of the pipeline kept frozen. We organise this section accordingly, addressing each of the three components in turn, followed by two cross-cutting techniques that recur across all three targets.

6.1 Trained Retrieval Components

Off-the-shelf encoders such as ColPali [61] score pages in isolation, leaving three gaps that retriever training aims to address. Modality-alignment variants pretrain a visual document encoder against OCR text: VDocRAG [47] compresses page images into single-vector embeddings while preserving alignment with their OCR-derived textual content, retaining text alignment without the storage and latency cost of late-interaction patch embeddings. Cross-page-fusion variants freeze the encoder and learn a neighbour-blending head: DFVC [34] adds a multi-layer perceptron adapter (using LoRA [65], a low-rank parameter-efficient tuning method that adds small trainable matrices to frozen weights) with gating and residual connections that fuses each page embedding with its neighbours, so evidence spread across adjacent pages becomes retrievable as a unit. Logical-relevance variants train a small scorer over a page graph: MoLoRAG [36] fine-tunes a frozen Qwen2.5-VL scorer to rank neighbour pages during graph traversal, turning retrieval into a learned multi-hop walk rather than a single-step similarity match.

6.2 Trained Reasoning Components

Prompted reasoning leaves quality sensitive to prompt design and base-model behaviour, motivating approaches that train the reasoning component directly. Reward-driven variants train a reasoning policy with explicit evidence supervision. DocR1 [28] fine-tunes a vision-language backbone with Evidence-Page-Guided GRPO, a variant of Group Relative Policy Optimisation (a reinforcement-learning method that compares grouped rollouts to compute relative rewards) whose composite reward combines format compliance, answer accuracy, and a page-citation indicator that explicitly credits trajectories citing the correct evidence pages. Trace-imitation variants instil a reasoning chain through supervised fine-tuning and preference optimisation: Chain-of-Reading [31] bakes in a four-stage plan-execute-integrate-verify chain through a staged pipeline of LoRA supervised tuning, full-parameter tuning, and direct preference optimisation (DPO, which trains on pairs ranked by preference rather than scalar rewards) on pairs that prefer faithful chains over hallucinated shortcuts. Uncertainty-driven variants train an aggregation head rather than a trajectory: DocSLM [30] learns an entropy-based calibrator that selects the lowest-uncertainty answer across streaming page segments, providing a self-consistency signal that scales to long documents under tight parameter budgets without requiring full-document context.

6.3 Trained Agentic Orchestration

Prompt-only orchestration is unreliable on smaller open-weight backbones that cannot follow ReAct protocols. Behavioural-cloning variants distil agent trajectories from a stronger teacher: DocDancer [53] fine-tunes a Qwen3 agent on synthesised exploration-then-synthesis traces that pair `Search` and `Read` ReAct actions with intent-conditioned tool selection, training the smaller model to match the trajectory shape of a larger teacher rather than learning the underlying policy from scratch. Reinforcement-learned variants reward correct tool use and trajectory shape: Doc-V* [60] couples supervised fine-tuning on coarse-to-fine ReAct traces with GRPO over a two-action space of retrieval and fetch calls plus entropy-based stopping that learns when to commit to an answer, while MACT [50] trains a multi-agent policy with agent-wise adaptive test-time scaling that allocates sample budgets across the planner, executor, judge, and answerer roles per question, spending more samples on the role that the question's difficulty most stresses rather than running every agent at a fixed budget.

6.4 Cross-Cutting Training Techniques

Two techniques recur across retriever, reasoner, and orchestrator training, independent of which component is being optimised.

Parameter-Efficient Tuning. Full backbone fine-tuning is prohibitive at long-document context lengths and erases pretrained capabilities, so the dominant pattern freezes the backbone and trains lightweight modules through parameter-efficient tuning approaches such as LoRA, projection heads, or task-specific adapters. LayTokenLLM [20] freezes the language model and adds a layout tokenisation module plus LoRA, encoding spatial layout through compressed positional tokens with negligible context overhead. CREAM [38] applies the LLaMA-Adapter-V2 recipe above a frozen Pix2Struct encoder, training only LoRA matrices, attention pooling, and bias and normalisation parameters in its answerer. The same pattern appears in DFVC's frozen-ColPali adapter and DocDancer's LoRA trajectory tuning, confirming that the choice between parameter-efficient and full fine-tuning is more often a compute-budget decision than a target-component one.

Multi-Stage Curriculum Training. Several systems converge on multi-stage curricula that start on single-page or local objectives before introducing document-scale supervision, exploiting the fact that text reading and layout understanding transfer from single-page data while cross-page integration must be learned separately. mPLUG-DocOwl2 [24] uses three stages of single-image structure learning on DocStruct4M, MP-DocStruct1M continued pretraining, and multi-task fine-tuning. DocSLM [30] extends this to a five-stage schedule that adds image-OCR alignment at the start and streaming-abstention calibration at the end.

7 Datasets

Underpinning the retrieval-augmented and iterative pipelines surveyed above is a dataset ecosystem strained by the cost of annotating long documents and by the field's reliance on synthetic generation to fill that gap. Datasets here span three roles: pretraining, fine-tuning, and benchmarking. Pretraining historically relied on single-page archives such as IIT-CDIP [90] and OCR-IDL [91], with genuinely multi-page corpora only recently emerging through DocStruct4M's mixed single- and multi-page pretraining [2] and the purely multi-page MP-DocStruct1M, Doc-750K, and PaperPDF [24; 32; 45]. Fine-tuning has shifted toward bespoke synthetic generation per model, with strong vision-language models writing question-answer or trajectory data over scraped PDFs since per-page annotation does not scale to long

Dataset	Pages	Type	Venue	Domain	# Docs	Avg. Pages	# Q/A	M/S Hop
DocVQA (2021)	SP	PT+BM	WACV	Industry	12767	–	50,000	S
InfographicVQA (2022)	SP	PT+BM	WACV	Infographics	5,485	–	30,035	M+S
ChartQA (2022)	SP	BM	ACL	Charts	20882	–	32,719	M+S
TextVQA (2019)	SP	PT+BM	CVPR	Scene-text images	28408	–	45,336	S
VisualMRC (2021)	SP	PT+BM	AAAI	Webpages	10197	–	30,562	M+S
WikiTableQuestions (2015)	SP	PT+BM	ACL	Wikipedia tables	2108	–	22,033	M
TabFact (2020)	SP	PT+BM	ICLR	Wikipedia tables	16573	–	118,275	M+S
FeTaQA (2022)	SP	PT+BM	TACL	Wikipedia tables	10330	–	10,330	M
PDF-VQA Task A (2023)	SP	PT+BM	ECML-PKDD	Biomedical	–	–	81085	S
PDF-VQA Task B (2023)	SP	PT+BM	ECML-PKDD	Biomedical	–	–	53872	S
Doc-750K (2025)	MP	IT/FT	CVPR	Academic Paper	–	–	758,000	M+S
DUDE (2023)	MP	BM	ICCV	Cross-domain	5019	5.72	41,541	M+S
LongDocURL (2025)	MP	BM	ACL	Cross-domain	396	85.6	2,325	M+S
M-LongDoc (2025)	MP	PT+BM	EMNLP	Cross-domain	180	210.8	851	S
M3DocVQA (2025)	MP	BM	CVPR	Wikipedia Articles	3368	12.2	2,441	M+S
MM-NIAH (2024)	MP	BM	NeurIPS	Synthetic	–	–	16,486	M+S
MMLongBench-Doc (2024)	MP	BM	NeurIPS	Cross-domain	135	47.5	1,082	M+S
MMVQA (2024)	MP	PT+BM	IJCAI	Biomedical	3146	9.6	262,928	M+S
MP-DocVQA (2023)	MP	PT+BM	PR	Industry	5928	8.27	46,176	S
SlideVQA (2023)	MP	PT+BM	AAAI	Presentations	2619	20	14,484	M+S
UDA-Benchmark (2024)	MP	BM	NeurIPS	Cross-domain	2965	–	29,590	M+S
DuReaderVis (2022)	MP	PT+BM	ACL	Cross-domain	158000	–	15,000	S
QASPER (2021)	MP	PT+BM	NAACL	Academic Paper	1585	~8	5,049	M
Kleister-NDA (2021)	MP	PT+BM	ICDAR	Legal	540	5.98	2,160	S
Kleister-Charity (2021)	MP	PT+BM	ICDAR	Financial	2778	22.19	21,612	S
MultiHiertt (2022)	MP	PT+BM	ACL	Financial reports	2513	~2.5	10,440	M
DocBench (2024)	MP	BM	preprint	Cross-domain	229	–	1,102	S
DocGenome (2024)	MP	PT+BM	NeurIPS	Academic Paper	500000	13	3000	M+S
MMLongBench (2025)	MP	BM	NeurIPS	Cross-domain	–	–	13,331	M+S
BRIDGE (2026)	MP	BM	preprint	Academic Paper	262	–	11,857	M
PDF-VQA Task C (2023)	MP	PT+BM	ECML-PKDD	Biomedical	1147	~10	5653	M
InstructDoc (2024)	MP+SP	IT/FT	AAAI	Cross-domain	–	–	–	M+S
TAT-DQA (2022)	MP+SP	PT+BM	ACM	Financial reports	2758	1.33	16,558	M
DocStruct4M (2024)	MP+SP	PT+IT/FT	EMNLP	Cross-domain	–	–	4,000,000	M+S

Table 2: Datasets referenced in MP-VRDU models. **Pages**: document scope (MP = Multi-Page, SP = Single-Page, MP+SP = Mixed). **Type**: dataset role – PT (Pretraining), IT/FT (Instruction Tuning / Fine-Tuning), BM (Benchmark), and combinations. **Hop**: question-hop type (Single, Multi, or both). '–' indicates not reported.

documents [22; 54; 28; 53]. Benchmarks have evolved from mid-length extractive evaluation on MP-DocVQA, DUDE, and SlideVQA [1; 92; 79] to long-context multimodal evaluation on MMLongBench-Doc, LongDocURL, and DocBench [93; 73; 85], with retrieval and page-localisation increasingly scored alongside answer accuracy [80; 28; 51]. Five benchmarks (MP-DocVQA, MMLongBench-Doc, DUDE, DocVQA, InfographicVQA [4]) appear in roughly half of recent MP-VRDU papers and constitute the de-facto standards for cross-model comparison (revisited in §8.1). Per-dataset statistics appear in Table 2.

8 Open Gaps

The preceding sections have organised MP-VRDU through four architectural families, three modality-specific challenges, the inference-time compositions and training recipes that recur across them, and the dataset ecosystem that supports them. Read together, these strands surface three persistent gaps that cut across the architectural taxonomy and remain unresolved by any current approach.

8.1 Data Scarcity and Evaluation Integrity

The first gap concerns the data foundation on which MP-VRDU progress is measured. Per-page annotation does not scale to documents averaging eight to twenty pages, leaving most systems to rely on vision-language-model-synthesised question-answer pairs and trajectories generated over scraped PDFs [22; 54; 28; 53]. This pragmatic response to scarcity has produced a structural integrity problem. Of the twelve papers in this review that introduce a synthesised corpus, nine source their PDFs from existing evaluation benchmarks, with MP-DocVQA, DUDE, and DocVQA appearing in both the fine-tuning mixes and the evaluation suites of large fractions of the surveyed papers. Performance reported as zero-shot therefore routinely includes the test corpus in its training history, with models trained on a strong vision-language model's behaviour over benchmark PDFs and then evaluated against those same PDFs under metrics often judged by the same model family. Compounding this, no paper in the surveyed corpus reports synthetic-data quality audits, document-level de-duplication against test pools, or backbone-level contamination checks, while agentic stochasticity and per-paper benchmark curations block direct cross-system comparison even when contamination is absent. Dominant benchmarks remain overwhelmingly English-centric, leaving non-English layouts, scripts, and document conventions underrepresented. The cumulative effect is that headline numbers in the surveyed literature cannot be taken as faithful indicators of progress, and the field lacks the evaluation infrastructure required to distinguish genuine capability gains from contamination or synthetic-evaluation artefacts.

8.2 The Evidence Pipeline Bottleneck

The second gap concerns a constraint that no architectural family has overcome: MP-VRDU systems must reconcile high-resolution per-page reading with document-scale evidence aggregation, an objective that exceeds any MLLM's compute and context budget [93]. The bottleneck is fundamental rather than incidental. Across the four families of §3, none has decoupled answer quality from the coverage-fidelity trade-off introduced in §4: page-to-document encoders suffer compression loss, long-context backbones suffer attention dilution, retriever-generator pipelines suffer a single-pass cutoff, and agentic systems inherit the recall of every tool they call. Current architectures merely shift where in the pipeline the trade-off is taken. Compression is also typically fixed at index time and independent of the question, so evidence is discarded before the question is seen, and chunking errors, OCR artefacts, parser failures, and per-passage scoring that ignores cross-references propagate directly into the answer. Iterative retrieval lifts this ceiling partially but inherits the recall of every tool it calls, so each additional round defers rather than dissolves the bottleneck.

8.3 Cross-Page Structural Reasoning

The third gap concerns a problem distinctive to the multi-page setting that single-page mechanisms cannot address. Cross-page structural reasoning currently sits between three imperfect routes (§4.3), none of which addresses the problem end-to-end at the cost and reliability that production MP-VRDU systems would require. Layout-as-objective training [24; 32] demands document-scale structural supervision that is expensive to collect at pretraining volumes. Implicit pixel-based recovery depends on backbone capability and high-resolution input, so the same compression pressures discussed in §4.2 weaken the structural signal whenever pages are compressed to fit document-scale inputs. Explicit parsed structure inherits the parser fragility detailed in §4.3, and the parsers themselves rarely recover relationships such as theorem-proof or footnote-reference linkages that cross page boundaries. The result is that each route trades a different cost: supervision, reliability under compression, or parser fidelity. No current approach offers a route that is simultaneously affordable to train, robust under document-scale inputs, and accurate on heterogeneous long-form documents.

9 Future Research Directions

The three gaps in Section 8 point toward several concrete directions the MP-VRDU community can pursue.

Addressing data scarcity and evaluation integrity requires investment in evaluation infrastructure that has so far lagged behind model development. Held-out closed-test benchmarks that are not used in any pretraining or fine-tuning corpus would establish baseline cross-system comparability, while mandatory contamination declarations covering pretraining, fine-tuning, and backbone leakage would allow the field to distinguish genuine generalisation from artefacts of overlapping corpora. Generator-evaluator separation, in which the model that synthesises training data is structurally prevented from also judging evaluation outputs, would break the current circular evaluation regime. Trajectory-level metrics paired with retrieval and grounding scores, building on the evidence-aware evaluation seen in MMLongBench-Doc [93], LongDocURL [73], and recent reasoning-augmented systems [28; 51], would surface the localisation failures that answer-only accuracy masks. Multilingual corpora covering non-English layouts, scripts, and document conventions would extend the field's reach beyond its current English-centric scope. The rapid succession of architectural families also makes cross-family comparison difficult to draw without shared evaluation protocols and consistent terminology, and progress on any of the above would benefit from broader coordination on reporting standards.

Addressing the evidence pipeline bottleneck requires rethinking compression and retrieval as question-dependent rather than question-independent operations. Compression that retains evidence based on the question's evidential demands, rather than discarding fine-grained content at index time, would push the trade-off later in the pipeline where it can adapt to the query. Retrieval that follows cross-references and logical relations rather than scoring passages in isolation would address the multi-hop and aggregation questions that single-pass similarity retrieval currently fails on, building on the early graph-traversal and hierarchical-index work surveyed in §5.1 but extending it toward retrievers that can plan over document structure rather than reacting to it. Methods that combine these properties may offer a path toward decoupling answer quality from the coverage-fidelity trade-off rather than merely shifting it.

Addressing cross-page structural reasoning requires advances on both the supervision and representation fronts. Cheaper structural supervision through weakly supervised approaches could mitigate the annotation cost that currently limits layout-as-objective training. Learned cross-page representations that survive compression would address the implicit-recovery route's dependence on high-resolution input. End-to-end structure-aware retrieval that does not depend on offline parser output would close the loop on the parser-fragility problem, treating structure inference as a learned component of the retrieval pipeline rather than a brittle preprocessing step. Across all three directions, progress will depend on benchmarks that explicitly evaluate cross-page structural reasoning rather than treating it as an incidental dimension of answer accuracy.

LLM Usage and Other Notes

LLM Usage. Large language models, specifically Claude Opus 4.7 and GPT 5.5, were used during the preparation of this review as a writing aid for spelling and grammar correction, sentence-level rewording, and refinement of reading flow. The taxonomy, argumentation, paper selection, and all claims about specific systems are the author's own, and every citation, factual statement, and quantitative figure was verified against primary sources before inclusion. Intellectual responsibility for the final document rests with the author.

Tables and Figures. All tables in this review were produced by the author. The source code used to generate them, along with the full revision history of this document, is available in the project repository linked under Supporting Materials below. Figure 1 was produced by the author in powerpoint, with final styling and formatting adjustments done by Dr. Yihao Ding.

Relationship to a Companion Survey Paper. Sections 3–8 are adapted from a companion survey paper on the MP-VRDU frontier of which I am first author. The underlying material was drafted largely by me prior to co-author contributions to that survey; the version presented here trims and re-tones the original prose and reorients the discussion toward the methodology directions motivated in §§8–9 rather than the broader scope of the survey itself.

Supporting Materials. The following supporting artefacts were produced during the preparation of this review:

- Paper inventory and cross-reference spreadsheet (surveyed systems, dataset and benchmark coverage, evaluation metrics):
  `https://docs.google.com/spreadsheets/d/1ZoPeIvnZKkHOQN8G3Fjj9gY8jnLImZvcYQrL36njS78/edit?usp=sharing`
- Project repository (literature review LaTeX source, build script, bibliography, and supporting code):
  `https://github.com/LeweiXu/MP-VRDU/`
- Companion survey paper will be available on arXiv in early June, named *A Survey on the Evolving Frontier of Multi-Page Visually Rich Document Understanding: Architectures, Trends, Methods, and Challenges*.

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